

The Liquidity Premium of Government Debt under a Floor System of Monetary Policy

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Abstract

Investors derive non-pecuniary benefits from the liquidity of short-term government debt. However, the central bank affects the supply of reserves, which work as close substitutes. This paper contends that a floor system of monetary policy, where the central bank attains its monetary policy rate target without altering the reserves supply, allows for untangling the role of fiscal and monetary policies in determining the liquidity premium on short-term government debt. Focusing on New Zealand's implementation of a floor system in July 2006, findings indicate that government debt issuance consistently reduces the liquidity premium, irrespective of its substitutability with other liquid assets. Additionally, unlike under a corridor system, monetary policy tightening within a floor system lowers the liquidity premium, adding to the negative impact on the government's fiscal capacity. Finally, under a floor system, the liquidity premium's cyclical dynamics may hinder the transmission of the monetary policy rate to rates governing consumption.

Key words: liquidity premium, monetary policy, treasury bills.

JEL codes: E43, E52, G12.

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1 Introduction

Investors derive non-pecuniary benefits from the liquidity of short-term government debt, rendering public debt a near-money asset that carries a liquidity premium, often referred to as a “convenience yield.” This phenomenon increases the price of government bonds, thereby expanding the government’s fiscal capacity. However, the central bank exerts influence over the supply of reserves and private deposits, which function as close liquid substitutes. This raises questions about what determines the liquidity premium on government debt. What is the role of the different monetary policy instruments, i.e., the policy rate and central bank reserves?

This paper argues that the best setting to answer these questions is one where the central bank implements monetary policy through a floor system of abundant reserves, not a corridor system (scarce reserves). In a corridor system, the central bank targets a positive spread between the interbank rate (at which banks borrow/lend reserves) and the interest paid on its reserves (IOR) and elastically supplies reserves to achieve that spread. Since the central bank effectively targets two interest rates, the quantity of reserves is not an independent policy instrument. Moreover, along the business cycle, central banks tightening monetary policy and draining reserves coincides with the government issuing fewer short-term T-bills. Or, if there is an increase in debt supply, the central bank needs to offset it with the opposite change in reserves to keep overall liquidity on target. Thus, it is challenging to observe the effect of debt supply in the data because the central bank also changes its supply of reserves to hit the interest rate target.

To disentangle the effect of debt supply, interest rates, and reserve supply, then you need (i) a setting where the policy interest rate is independent of reserve supply, and (ii) to isolate the effect of debt supply, a setting where the central bank does not adjust its reserve supply to hit its target. A floor system provides both: by supplying abundant reserves to the extent that banks’ demand for liquidity no longer responds to the quantity of reserves, the interbank rate drops to equal the IOR. Since reserves are no longer supplied elastically to target two interest rates, the IOR can be set independently from the reserve supply. In addition, the overall supply of liquidity (reserves plus near-substitutes such as T-bills) no longer affects the stance of the interest rate, and thus the central bank does not need to offset new debt issuance by the government.

Although the U.S. Federal Reserve has conducted something close to a floor system since late 2008 (when it increased the reserve supply and started paying interest on them), the Quantitative Easing (QE) programs deployed in 2009-2013 endogenously altered the supply

of reserves, and the IOR remained unchanged until December 2015. For this reason, the empirical exercise is implemented using data from New Zealand, which transitioned to a floor system in 2006 and did not reach the zero lower bound or deploy QE until March 2020. The analysis reveals that the liquidity premium decreases with the debt supply, validating the role of government borrowing. Surprisingly, the level of the IOR substantially reduces the liquidity premium. This finding contrasts with Nagel’s (2016) positive effect of the interbank rate. Under a corridor system with no interest paid on reserves, the interbank rate represents the opportunity cost banks face in holding liquidity in the form of reserves. A higher interbank rate increases the opportunity cost of this form of liquidity, raising the liquidity premium of substitutes such as government debt. However, under a floor system where the IOR equals the interbank rate, this opportunity cost is zero, and one would not expect it to impact the liquidity premium of government debt¹.

To explain these results, I propose a monetary model with two key additions. First, deposits issued by banks provide transaction services to households. This is captured by introducing deposits into the household’s utility function. Therefore, deposits are a cheaper source of funding for banks. Still, when they issue deposits, they encounter two frictions: (1) liquidity shocks, which can only be managed by holding reserves, and (2) leverage constraints: to back deposits, banks can invest in high-quality assets (central bank reserves or short-term government bonds). Thus, central bank reserves and short-term government bonds earn a convenience yield equal to the difference between the illiquid rate (on loans, which governs household consumption/borrowing decisions) and their yields. From the banks’ perspective, the liquidity premium thus represents the cost of collateral.

The central bank supplies abundant reserves in a floor system, rendering them irrelevant for managing liquidity shocks (although they still work as collateral). The interbank rate falls to equal the IOR, disentangling the supply of reserves from the interest paid on them. Consequently, to fulfill its inflation mandate, the central bank must only set the IOR. This has two noteworthy consequences: first, from the private bank’s perspective, the only difference between reserves and T-bills lies in collateral quality. Second, notice that although the reserve supply does not need to be closely monitored to achieve the policy rate target, it still matters as it affects the supply of collateral available to banks.

¹Nagel (2016) discusses the effect of the *introduction* of reserves remuneration on the liquidity premium in the UK and Canada (within a corridor system), and finds no effect. However, both the empirics and the model are not fully comparable. Modelwise, Nagel (2016) acknowledges that his framework is not suitable to analyze an abundant-reserves system (the details are explained later). Empirics-wise, I test the effect of the IOR when it explicitly acts as the monetary policy rate.

The second key addition is that the central bank does not control the short rate of the household’s Euler equation (the nominal rate in the Fisher relationship). Instead, the policy rate corresponds to the rate paid to banks on their reserves. It differs from the rate governing intertemporal consumption by an amount equal to the liquidity premium on central bank reserves. This feature will be necessary to explain why, under a floor system where the opportunity cost of holding reserves is zero, a monetary policy tightening decreases the convenience yield on liquid assets, including T-bills. This contrasts with the models of Nagel (2016) and Drechsler, Savov, and Schnabl (2017), where the monetary policy rate coincides with the illiquid nominal rate governing intertemporal consumption, and any exogenous monetary policy tightening will mechanically increase the convenience yield because it increases the illiquid rate. Also, it cannot address a floor system because the significant increase in reserve supply needed to implement it would imply that the IOR and the T-bill yield rise to equal the monetary policy (illiquid) rate, and the liquidity premium would always be zero for all assets.

The model is then employed to illustrate how monetary and fiscal policy interact in determining the liquidity premium of government debt. Three novel results emerge: first, increasing government debt issuance reduces the liquidity premium regardless of its substitutability with other liquid assets. Second, exogenous monetary policy tightenings under a floor system negatively impact the government’s fiscal capacity through their adverse effect on the liquidity premium. The key mechanism works through the marginal valuation of liquid assets. Banks shift their marginal valuation toward reserves, as they are now being paid a higher yield for this type of high-quality collateral. Therefore, their cost of collateral drops, reducing the marginal value of liquidity services. This lowers the willingness to pay for the liquidity provided by T-bills. This is the opposite effect under a corridor system. Third, under a floor system, the cyclical dynamics of the convenience yield dampen the transmission from the IOR to the intertemporal rate, impairing monetary policy transmission. These findings bear implications for all central banks expected to maintain large balance sheets in the future.

Literature review. The determination of convenience yields is relevant for fiscal and monetary policy. Convenience yields lower short-term equilibrium interest rates (Del Negro et al., 2019; Lenel, Piazzesi, and Schneider, 2019), are at the center of unconventional monetary policy transmission in Krishnamurthy and Vissing-Jorgensen (2011) and Del Negro et al. (2017) and expand the government’s fiscal capacity (Mian, Sufi, and Straub, 2021; Reis, 2021; Jiang et al., 2022).

This paper contributes to the empirical literature on convenience yields by untangling the roles of the supply of government debt and monetary policy. The early papers on convenience yields of short-term government debt usually ignore the role of monetary policy (Krishnamurthy and Vissing-Jorgensen, 2012; Greenwood et al., 2015). Building upon the work of Nagel (2016) and d’Avernas and Vandeweyer (2024), I extend the analysis by exploring the impact of interest rates and the supply of reserves under a floor system.

Additionally, I show that, unlike under a corridor system (Nagel, 2016), under an abundant reserve system the supply of T-bills has a negative impact on the liquidity premium. Krishnamurthy and Li (2023) extend Nagel (2016) in the direction of including a broader definition of private liquid assets (beyond only bank deposits). They also find a significant role for debt supply in explaining the liquidity premium of government debt, but their explanation rests on its lower degree of substitutability with financial sector debt. My explanation is complementary to theirs, as they do not study the liquidity premium under different monetary policy frameworks. Cuevas (2025) also finds a negative impact of debt supply on the liquidity premium, but the frameworks are different. That paper studies the mechanisms behind the link between the policy rate and the liquidity premium when debt issuance respond endogenously to monetary policy, regardless of the monetary framework².

The model presented in this paper extends a framework familiar to other studies where banks derive a convenience yield from holding bonds. While Lenel, Piazzesi, and Schneider (2019) argue that bonds alleviate banks’ balance sheet costs, I simplify this by assuming a reduced-form collateral constraint. The modeling of the floor system in this paper aligns closely with Piazzesi, Rogers, and Schneider (2021). However, we differ in that they highlight the role of nominal rigidities, they exclude a government, do not explore fiscal and monetary policy interactions, and do not explore the role of liquidity demand shocks. They focus on optimal interest rate policy in a standard New Keynesian model with banking.

Related studies analyze the convenience yield of bank deposits and their role in monetary policy transmission (Dreschler, Savov, and Schnabl, 2017, 2018), highlighting well-known market structure issues in the banking sector, such as market power in the market for bank deposits. My results do not aim to challenge and are compatible with these results. For the sake for simplicity, I streamline the dynamics of liquid private deposits to maintain focus on the convenience yield of government debt.

Floor systems have been extensively studied in New Keynesian models, primarily fo-

²The broader implications of supply and demand effects in the Treasury market are central to recent papers, such as Greenwood and Vayanos (2010, 2014), Krishnamurthy and Vissing-Jorgensen (2011), Hanson and Stein (2015), and Vayanos and Vila (2021).

cusing on optimal policy and their impact on real activity (Curdia and Woodford, 2011; Canzoneri, Cumby, and Diba, 2017; Arce et al., 2020; Piazzesi, Rogers, and Schneider, 2021; Benigno and Benigno, 2021). In contrast, this paper aims to describe the interaction between the monetary and fiscal authorities and their combined impact on fiscal sustainability and monetary policy transmission.

The paper’s structure unfolds as follows: Section 2 explains the empirical strategy and its results. Section 3 presents a stripped version of the model that highlights the mechanism through which the monetary policy rate impacts the liquidity premium under a floor system. Section 4 presents the full model, delving into how monetary and fiscal policy interact to determine the liquidity premium of government debt throughout the business cycle. Section 5 provides concluding remarks.

2 Empirical analysis

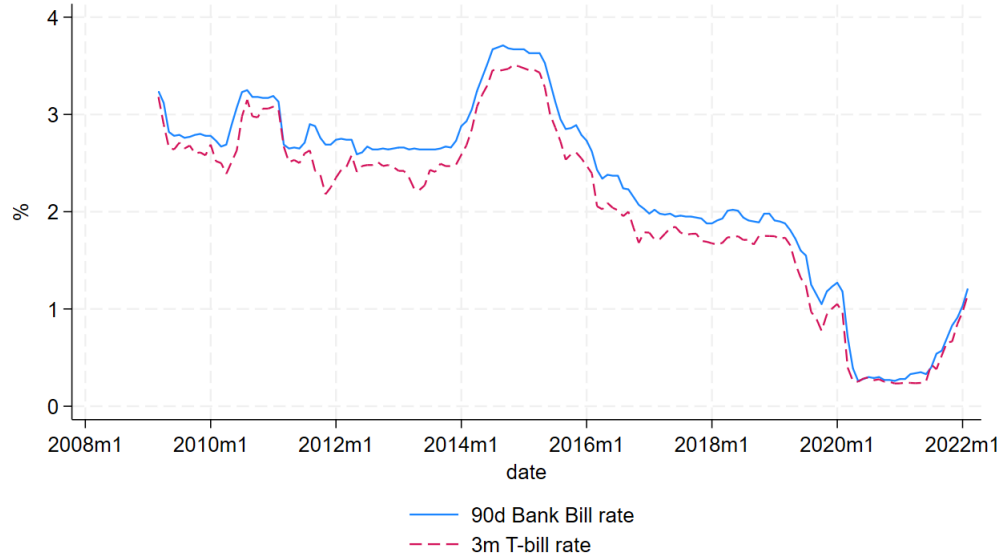
2.1 Data

I follow the literature’s convention and measure the liquidity premium of T-bills by taking their spread against other assets of similar maturity and credit risk. The intuition is that since the maturity and credit risk are the same, the remaining spread must be due only to their liquidity services. The spread, then, is a measure of the extra price investors are willing to pay for the liquidity of the T-bills. For New Zealand data, I use two assets comparable to T-bills: 3-month bank bills and 6-month term deposits. Appendix A describes all data sources.

Bank bills are a typical investment in Australia and New Zealand: it is a promise by the borrower to pay the face value at maturity. It is considered a safe investment because they are “accepted” (guaranteed) by a bank. In accepting the bank bill, the bank makes the payment at maturity, regardless of the borrower’s repayment ability. Investors can buy bank bills from a bank for an agreed face value, an agreed term, and a quoted interest rate. The bank then endorses the bank bills to acknowledge the change in ownership. Upon maturity, the bank will pay the total face value, which includes the initial purchase price and the interest receivable. An additional feature of investing in bank bills is that should you require funds before the due date, the bank will purchase the bank bill at the prevailing interest rate. Although this last feature might make them as liquid as a T-bill, they have consistently traded at yields above T-bills, with the spread spiking during the turmoil of 2008.

On the other hand, the spread between term deposit rates and T-bill rates is one of the

Figure 1: 90-day Bank Bill and 3-month T-bill



Notes: Figure shows the 90-day Bank Bill and 3-month T-bill from March 2009 to 2022. The spread captures the liquidity premium.

measures used in Krishnamurthy and Vissing-Jorgensen (2012) for U.S. data. Like the U.S., term deposits in New Zealand are a safe investment where investors lock in a competitive interest rate for a fixed term. Investors are usually not allowed to redeem their money before the term's expiration. Some banks allow it in specific circumstances and require early notifications. Therefore, the spread should account for the liquidity provided by the T-bill.

The liquidity premium measured with these two assets is sizable. The bank bill yields 28 basis points above T-bills on average for January 1996-July 2021. Figure 1 shows the evolution of the bank bill and T-bill rates, both at the 3-month maturity. The spread captures the liquidity premium. The term deposit has a larger spread of 63 basis points on average. The larger spread for the term deposit might come from two sources. First, its maturity is six months (both legs at six months) rather than three. Second, this might be due to the higher liquidity of the bank bill, which makes it more similar to the T-bill. Recall that the bank bill can be sold any time during the instrument's term, while the investment in the term deposit is locked in for the whole period. Standard deviations are high: 24 and 80 basis points, respectively.

2.2 Monetary Policy Frameworks and Liquidity Premia

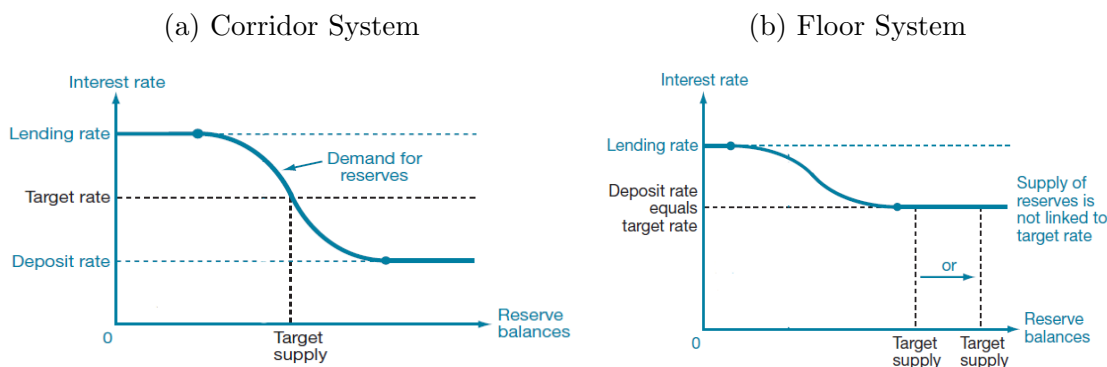
The liquidity premium can be priced by households or by financial institutions. Households can derive non-pecuniary liquidity services from deposits at the bank or from T-bills (if they access them via money market funds, for example). Financial institutions, on the other hand, can enjoy non-pecuniary liquidity services by holding reserves at the central bank or by holding T-bills. Therefore, depending on the level of the analysis, T-bills could be a close substitute for deposits (for households) or central bank reserves and other liquid assets (for banks). In most banking models, the supply of bank deposits is closely linked to the supply of central bank reserves (for example, via reserve requirements), and monetary policy decisions that alter the supply of reserves also translate into changes in the supply of deposits. In this Section, I will show correlations between the supply of T-bills and both deposits and central bank reserves. In the model, the liquidity premium will arise at the bank level, but I will discuss the implications if they arise at the household level.

This Section provides evidence that under a floor system the following two facts hold: (i) the monetary policy interest rate is independent of reserve supply, and (ii) the central bank does not adjust its reserve supply to hit its target, untangling monetary policy from the supply of government debt.

Under a floor system, the central bank issues abundant reserves beyond the point at which banks' reserve demand is flat and does not respond to changes in supply. This can be understood through the banks' reserve demand model in Poole (1968) or Afonso et al. (2020). In these models, banks demand reserves only to meet reserve requirements, and there are no frictions in the interbank market. Demand is flat at high levels of aggregate reserves, with banks indifferent among a wide range of reserve holdings as long as market rates (including the interbank rate) equal the interest rate paid on reserves (see Figure 2b). Since, in this case, central banks operate on the flat part of the banks' demand for reserves, they, in practice, have two independent instruments: the policy rate (the interest on reserves) and the supply of reserves. The former is used to meet its mandate on inflation, and the latter ceases to be closely monitored. In addition, the overall supply of liquidity (reserves plus near-substitutes such as T-bills) no longer affects the stance of the interest rate, and thus the central bank does not need to offset new debt issuance by the government.

The U.S. Federal Reserve has implemented a system remarkably similar to a floor system since October 2008, when it expanded the supply of reserves and began paying interest on them. However, since then, this policy has mostly coincided with the conduct of QE (where reserve supply becomes endogenous) and didn't alter the IOR for seven years. Therefore, I

Figure 2: Two Systems of Monetary Policy Implementation



Note: Figures taken from Keister, Martin, and McAndrews (2008)

study the liquidity premium with data from New Zealand. In July 2006, the Reserve Bank of New Zealand (RBNZ) initiated a transition from a symmetric channel system to a floor system (see the Appendix B for a detailed description of this transition and its key elements). The level of reserves went from NZ\$20 million to around NZ\$8 billion in a few months. This level was kept steady until March 2020. At the same time, the interest on reserves (called the Official Cash Rate) was set equal to the interbank rate. Notably, New Zealand did not reach the zero lower bound nor conducted QE until the COVID crisis in March 2020 (when the reserves supply was raised to almost NZ\$30 billion), which makes it an ideal scenario to study a floor system.

Data for New Zealand supports the assumption that reserve supply is no longer endogenous to monetary policy under a floor system. In Figure 3a, there is no clear correlation between the interest on reserves and the supply of reserves. With the supply of deposits (not shown), the correlation is not significant (correlation of 0.096, p-value=0.30). This correlation would be significantly negative under a corridor system, such as the one in the United States before 2008. In Figure 3b, the correlation between T-bills and reserves is also not significantly different from zero (correlation of 0.007, p-value=0.943), showing the disconnection between the two under a floor system. The Appendix B looks more closely at these correlations and discusses a few episodes where the RBNZ injected liquidity into the banking system, most likely driven by liquidity concerns.

For comparison, let's look at the equivalent Figures for a period where monetary policy is implemented through a corridor system. In these time periods, data shows that T-bills and reserves or deposits positively correlate along the business cycle. In Figure 4a, I show the correlation between T-bills supply and reserves supply for the United States between

Figure 3: Correlations for New Zealand: Jul. 2006 - Feb. 2020

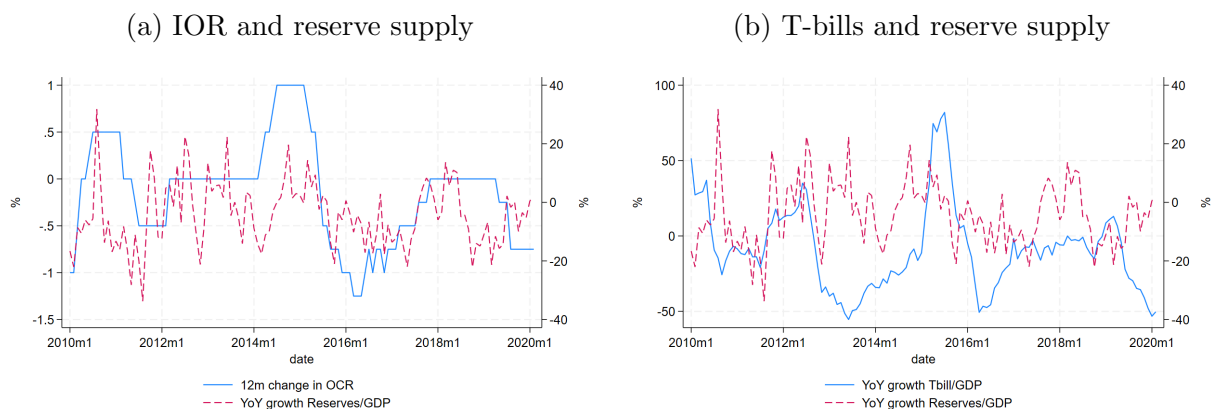
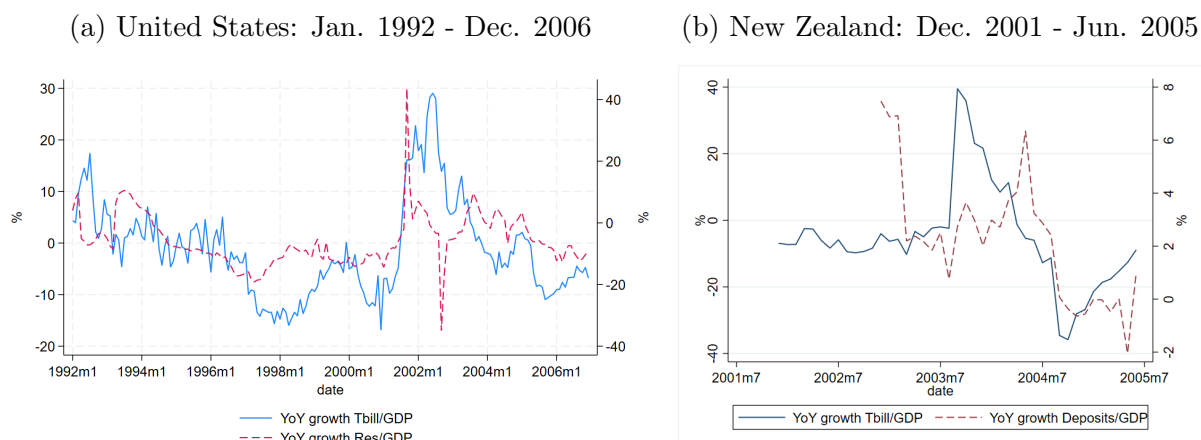


Figure 4: Correlation between T-bills and Reserves (Scarce-Reserve System)



1992 and 2006. The positive correlation is significant at the 1% level. The spike in Panel 4a corresponds to September 2001. This result is robust when using deposits-to-GDP instead of reserves. It is challenging to develop an equivalent Figure for when New Zealand implemented a corridor system, as during this period, banks relied primarily on intra-day repos rather than on central bank reserves. However, Panel 4b shows that a similar positive correlation arises between T-bills and private checkable deposits, which is an alternative liquid asset (significant at the 1% level, from December 2001 to June 2005, before the floor system).

What explains the positive correlation in Figure 4? The reason might be that the central bank responds to T-bill supply or that both the Treasury and the central bank respond to households' demand for liquidity. There could be macroeconomic shocks or long-term stochastic trends. It is challenging to disentangle a precise explanation, and I don't attempt

to do it here. The point of Figure 4 is only to show the collinearity between the two series³.

Yields on short-term Treasuries are low because Treasuries are very liquid. Increased Treasury supply by itself would lower the equilibrium price of liquidity. However, it is challenging to observe this effect in a corridor system due to the collinearity with the supply of reserves or deposits. Nagel (2016) shows that the federal funds rate exhibits variations at higher business cycle frequencies, and these higher-frequency movements are correlated with movements in the liquidity premium, resulting in a significantly higher R^2 . T-bill supply is less persistent and more highly correlated with the level of the short-term interest rate than total debt-to-GDP, but some of the movement in this variable does not align as well with the liquidity premium as the federal funds rate does. Thus, he concludes that once we take into account that the liquidity premium of near-money assets should be related to the opportunity cost of money (proxied by the federal funds rate), there is little evidence that Treasury supply has any incremental effect on the level of the T-bill liquidity premium.

If Treasury supply is to have a causal effect on liquidity spreads, then you need to look at a setting at which the Fed does not adjust its reserve supply to hit its target⁴. Figure 3b shows that a floor system provides an opportunity to analyze the liquidity premium in instances where the supply of reserves is untangled from the rate target and the supply of debt.

2.3 Results

I estimate the following regression via instrumental variables:

$$LP_t = \alpha_0 + \alpha_1 i_t^m + \alpha_2 (Tbill/GDP)_t + \alpha_3 X_t + v_t \quad (1)$$

where X_t is a set of controls. In this equation, coefficients will have a different interpretation as in Nagel (2016). α_1 captures the effect of the level of interest rates stripped from its effect through the supply of reserves. A monetary policy tightening (an increase in the IOR) no longer involves a decrease in the supply of reserves. What is the expected sign of $\hat{\alpha}_1$? If the central bank pays a higher yield on its reserves, banks will ask for a higher yield on T-bills to be willing to hold them. A higher yield on T-bills will be accommodated via a

³This positive collinearity is consistent with the results of Krishnamurthy and Vissing-Jorgensen (2015), who found that Treasury issuance crowds out less liquid private deposits, such as savings and time deposits, but does not crowd out the most liquid deposits. In fact, checkable deposits-to-GDP are positively correlated with Treasury supply-to-GDP.

⁴A second option (not undertaken in this paper), would be to look at instances where Treasury supply drives the interest rate target.

lower liquidity premium. Therefore, we should expect $\hat{\alpha}_1 < 0$. The coefficient $\hat{\alpha}_2$ will capture the causal effect of the supply of T-bills, no longer confounded by the collinearity with the supply of reserves.

I estimate Equation (1) via instrumental variables. The dependent variable is the spread between the 6-month term deposit and the 6-month T-bill. Data starts in March 2009 (the month the New Zealand Treasury started auctioning 6-month instruments). The independent variables are the interest on reserves, the VIX index -to account for demand shifts due to uncertainty-, the supply of government debt (which I proxy by using the supply of T-bills or the supply of debt of all maturities), and the supply of deposits or reserves. The latter accounts for the few instances where the RBNZ injected reserves for endogenous reasons. I use two measures of the supply of deposits: the total amount taken from the banking system's balance sheet and the total amount taken from the households' balance sheets. Newey-West standard errors correct for autocorrelation.

For robustness, I run the regressions in the Appendix D using the spread between the 90-day Bank Bill and the 3-month T-bill to measure the liquidity premium. It also shows robustness in estimating the regression in first-differences and dropping the COVID-19 pandemic months.

Instrumental variables for the monetary policy rate are needed since an unobserved liquidity demand shock (correlated with interest rates or the supply variables) may induce the central bank to lower interest rates in the same month, leading to reverse causality. This would result in a downwardly biased estimate of the interest rate coefficient.

I follow Piazzesi and Swanson (2008) in instrumenting the monetary policy interest rate and utilizing their risk-adjusted forecast from futures contracts. Futures contracts settle at the end of each month based on the average rate that prevails during that month. The futures price in months before the expiration month should be highly correlated with the average monetary policy rate during the expiration month. However, Piazzesi and Swanson (2008) show that futures prices include a non-negligible risk premium that can be correlated with the liquidity premium. For this reason, I use the price of the 1-month ahead future contract from two months prior to the expiration month to avoid a correlation between the instrument and the dependent variable.

Instruments for T-bill are nonlinear (squared and cubic) functions of lags of total debt-to-GDP ratio. As argued in Krishnamurthy and Vissing-Jorgensen (2012) and Greenwood et al. (2015), nonlinear functions of total debt to GDP exploit the positive correlation between total debt and its maturity structure. Similarly, instruments for deposits and reserves are

Table 1: Determinants of the liquidity premium - 2SLS

	(1)	(2)	(3)	(4)
ior_t	-0.197** (0.086)	-0.237*** (0.085)	-0.247*** (0.078)	-0.282*** (0.051)
$\log(\frac{Tbill}{GDP})_t$	-1.404*** (0.233)	-1.262*** (0.211)		
$\log(\frac{Debt}{GDP})_t$			-2.045*** (0.406)	-1.731*** (0.278)
$\log(\frac{HHDepts}{GDP})_t$	-5.695*** (1.014)			
$\log(\frac{Deps}{GDP})_t$		-6.466*** (1.067)	0.572 (1.293)	
$\log(\frac{Reserves}{GDP})_t$				-0.239 (0.198)
vir_t	0.026*** (0.008)	0.026*** (0.008)	-0.024*** (0.006)	-0.017** (0.009)
Constant	2.005** (0.923)	6.616*** (1.470)	1.512 (1.799)	1.706*** (0.529)
Instruments for:				
IOR	Y	Y	Y	Y
Tbill/Debt supply	Y	Y	Y	Y
Deposits/Reserves	Y	Y	Y	Y
Weak instruments test				
CD stat	6.66	6.71	6.60	8.22
SY critical value	5.56	5.56	5.56	5.56
Observations	149	149	149	149

Notes: Data are at monthly frequency. The dependent variable is the 6-month Term deposit/Tbill spread, measured in percentage units. $\log(\frac{Debt}{GDP})_t$ includes outstanding government debt of all maturities. $\log(\frac{HHDepts}{GDP})_t$ are total deposits in households' balance sheet. $\log(\frac{Deps}{GDP})_t$ are deposits in the monetary base. Instruments for T-bill and debt supply are nonlinear functions of lags of total debt-to-GDP ratio. Instruments for deposits and reserves are nonlinear functions of lags of T-bills to GDP. A CD stat greater than the Stock and Yogo critical value rejects weak instruments (with bias greater than 20% of OLS bias). Newey-West standard errors in parenthesis (3 lags).

nonlinear functions of lags of T-bills to GDP⁵.

Table 1 shows the results. Columns 1 and 2 use the supply of T-bills as a measure of the supply of government debt. The coefficient on T-bill supply is negative and significant. A decrease in T-bills/GDP from 0.135 (it's sample mean) to 0.10 (roughly a drop of half a standard deviation) increases the liquidity premium by 37 basis points. This result is robust to the measure of the supply of deposits used. The magnitude is lower than in Krishnamurthy and Vissing-Jorgensen (2012) but is much larger than the non-significant effect found by Nagel (2016) for the U.S. data.

⁵The contemporaneous nonlinear transformation of T-bills/GDP is not included to avoid violating the exclusion restriction.

Interestingly, the coefficient of the interest on reserves is significantly negative. In Column 2, an increase of 1 percentage point in the IOR is associated with a drop in the liquidity premium of 25 bp. This is the effect of the monetary policy rate stripped from its connection with the supply of central bank reserves. It represents the effect of tightening monetary policy under an ample reserve system. This is a complete switch of the sign in the impact of the monetary policy rate on the liquidity premium estimated in Nagel (2016) under a scarce-reserves system. The Appendix E shows that this negative coefficient also shows up for data in the U.S. after 2008, even though, in that case, the caveat of an endogenous supply of reserves still applies.

Columns 3 and 4 replicate the first two columns but include the supply of debt of all maturities as a proxy for the supply of government debt. Both the coefficient on the IOR and the negative effect of government debt are robust.

Notably, the instruments in Table 1 are strong, as shown by the relatively high Cragg-Donald statistics. A statistic greater than the critical values from Stock and Yogo (2005) -also shown- rejects the hypothesis of weak instruments (with bias greater than 20% of the OLS bias) at a significance level of 5%. The Appendix C reports first-stage regression results.

3 Mechanism

This Section highlights the key mechanism by which monetary policy affects the liquidity premium of government debt under a floor system. The following Section presents the full version of the model and analyzes all its qualitative implications. The primary simplification of this Section is that the model is expressed in real terms, focusing exclusively on the interaction between government bonds and alternative liquid assets. Other issues that could complicate the exposition of the mechanism, such as inflation or the contrast with the corridor system, are addressed in the next Section.

This is an endowment economy-representative agent model comprising households, who derive transaction services from inside money (bank deposits); banks, which provide deposits by accumulating central bank reserves and other assets; a fiscal authority that sets the supply of T-bills; and a central bank that conducts monetary policy. Output is fixed at $Y_t = Y$, meaning that monetary policy affects financial decisions but not aggregate production. Households adjust savings and borrowings, but not total economic activity.

3.1 Household Problem

Households maximize:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [u(C_t) + \chi \log(D_t)] \quad (2)$$

subject to the budget constraint:

$$\begin{aligned} C_t + k_t A_t - L_t + D_t &= D_{t-1}(1 + r_{t-1}^d) \\ &+ k_{t-1}(A_t + Y_t) - T_t - L_{t-1}(1 + r_{t-1}) + \Omega_t \end{aligned} \quad (3)$$

where C_t and D_t are real consumption and deposits, respectively, and χ is a parameter. The real utility of holding deposits is a reduced-form way to capture the benefits of holding liquid assets, such as facilitating unmodeled intra-period transactions. k_t denotes the share of the total endowment stream that the household owns at the end of period t , A_t is the price of the claim to the endowment stream, T_t denotes transfers/taxes from the government, r_t^d is the interest rate on deposits, L_t denotes loans, Ω_t is the flow of profits from banks to households, and r_t is the interest rate applicable to assets and transactions that do not produce a liquidity service flow.

The household first-order condition for loans gives the intertemporal Euler equation

$$1 = \beta \mathbb{E}_t \left[\frac{u_c(C_{t+1})}{u_c(C_t)} (1 + r_t) \right] \quad (4)$$

where u_c denotes the first derivative of $u(\cdot)$. The illiquid rate r_t in Equation (4) governs the household's intertemporal decisions; thus, this is the rate the monetary authority looks to affect. However, as explained below, the central bank's policy rate refers to the rate on banks' reserves, which is defined later.

The first-order condition for real deposit balances is:

$$\frac{\chi}{D_t} = u_c(C_t) \frac{r_t - r_t^d}{1 + r_t} \quad (5)$$

Equation (5) reflects households' valuation of inside money or deposits. Since households derive transaction services from deposits, they are willing to accept an interest rate on deposits r_t^d below the intertemporal rate. The spread $r_t - r_t^d$ represents what I will call the "household's cost of liquidity", reflecting the value of liquid assets for transactions. It is declining in real balances: the marginal benefit of payment instruments is decreasing in the overall quantity held.

3.2 Banks

The representative bank in this economy exists because households cannot access the central bank's reserves directly. The bank, therefore, issues liquid deposits, on which they promise to pay a riskless return r_t^d one period later, by either holding reserves at the central bank, M_t^d , that receive a yield of r_t^m or having other assets such as loans, L_t^s , or T-bills, B_t^h , on which they earn an interest rate of r_t and r_t^b , respectively.

In most models of banking, reserves play a unique role in liquidity management: from a representative bank facing real intermediation and liquidity costs (Lenel, Piazzesi, and Schneider, 2019; Curdia and Woodford, 2011) to a mass of banks facing idiosyncratic liquidity needs (Arce et al., 2020; Piazzesi, Rogers, and Schneider, 2021). When reserves are scarce, as in a corridor system, banks are willing to forgo a return (the spread between the interbank rate and the interest paid on reserves) to hold them, and the central bank targets an interbank rate by altering the supply of reserves.

In the current setting, to resemble a floor system, I follow Piazzesi, Rogers, and Schneider (2021) and assume that because reserves are abundant, they play no unique role in managing liquidity costs. However, reserves play an additional role as high-quality collateral because banks face a leverage constraint: banks need to invest in central bank reserves or T-bills to back deposits. From the bank's perspective, the primary difference between reserves and other assets is the quality of collateral.

It must be emphasized that while reserves play no unique role in liquidity management, households' demand for liquidity remains positive due to Equation (2), which assumes no satiation in deposit holdings.

Banks' cash flows at date t are given by:

$$M_{t-1}(1 + r_{t-1}^m) - M_t + B_{t-1}^h(1 + r_{t-1}^b) - B_t^h + L_{t-1}(1 + r_{t-1}) - L_t - D_{t-1}(1 + r_{t-1}^d) + D_t \quad (6)$$

Banks maximize the present value of this cash flow, discounted at the illiquid rate, r_t , which represents the household discount factor and, hence, the banks' cost of capital.

Banks can issue deposits only if they have sufficient collateral to back them:

$$D_t \leq M_t + \rho_b B_t^h \quad (7)$$

where ρ_b captures the idea that reserves could be better ($\rho_b < 1$) or worse collateral ($\rho_b > 1$) than other assets. A leverage constraint can be viewed as a simple way to model an increasing marginal cost of debt⁶.

⁶See Lenel, Piazzesi, and Schneider (2019) for a model that features increasing marginal cost of debt via more micro-founded balance sheet costs.

Deposits represent a cheaper funding source than equity because of the non-pecuniary services deposits provide to households. Without a leverage constraint, it would be optimal to fund the bank entirely with deposits. Therefore, the leverage constraint will always bind in equilibrium.

The first-order conditions for reserves, T-bills, and deposits are, respectively,

$$\frac{r_t - r_t^m}{1 + r_t} = \gamma_t \quad (8)$$

$$\frac{r_t - r_t^b}{1 + r_t} = \rho_b \gamma_t \quad (9)$$

$$\frac{r_t - r_t^d}{1 + r_t} = \gamma_t \quad (10)$$

where γ_t is the Lagrange multiplier on the leverage constraint. A binding leverage constraint induces a spread between the illiquid rate and the rate on T-bills, which corresponds to the convenience yield on T-bills. This liquidity premium represents what I will call the “bank’s cost of collateral”, and it captures that banks value T-bills not only for their interest rate but also for their collateral value, which allows them to issue deposits. Notice that these conditions imply $r_t^m = r_t^d$, i.e. perfect pass-through from the IOR to the deposit rate. This equality is relaxed once I add liquidity costs, Ψ_{dt} , in the full version of the model in the next Section.

3.3 Government

The government issues T-bills and finances spending through taxation. Its budget constraint is:

$$T_t + B_t^h = (1 + r_{t-1}^b)B_{t-1}^h + G_t \quad (11)$$

where G_t is government spending and T_t are lump-sum taxes. For simplicity, assume $G_t = 0$ in every period.

3.4 Central Bank and Monetary Policy

The central bank directly sets the interest on reserves (IOR):

$$r_t^m = (1 - \rho)\bar{r} + \rho r_{t-1}^m + \epsilon_t^r \quad (12)$$

where $\bar{r}$ is the steady state IOR, ρ is a smoothing parameter, and ϵ_t^r is the exogenous monetary policy shock. This Taylor rule is certainly unrealistic since it neither targets

inflation nor the output gap. However, this is a momentary simplification to highlight the central bank's effect on financial decisions and the liquidity premium.

Market clearing conditions imply that $Y_t = C_t$.

Proposition 1. *An exogenous increase in the IOR reduces the liquidity premium on T-bills.*

Proof: differentiating both sides of the FOC for the IOR: $dr_t - dr_t^m = (1 + dr_t)d\gamma_t$. The exogenous shock corresponds to $dr_t^m > 0$. Since r_t is pinned down by the real rate (in the Euler equation) and output is constant, we have that $dr_t = 0$ and, therefore, $d\gamma_t < 0$. From the FOC for T-bills, $d(r_t - r_t^b) = \rho_b(1 + dr_t)d\gamma_t$, and thus $d(r_t - r_t^b) < 0$. $\square$

The distinctive feature of the monetary policy shock under a floor system is that it involves a change in the IOR without a change in the supply of collateral (reserves). Although the total supply of deposits, reserves, and T-bills has not changed, the exogenous tightening affects their marginal valuation. Banks shift their marginal valuation toward reserves (they now offer similar liquidity but higher yield), reducing their reliance on T-bills. Since $\rho_b < 1$, this loosens the leverage constraint and thus reduces the liquidity premium on T-bills. Banks are now being paid more for the high-quality collateral they hold. Therefore, their cost of collateral drops, reducing the marginal value of liquidity services. The lower marginal value of liquidity services lowers the willingness to pay for the liquidity provided by T-bills.

4 Model

In this Section, I show the complete version of the model. This remains a monetary and financial model with no nominal rigidities, as I aim to focus on the financial mechanisms (although I compare the implications with those of a version featuring endogenous income and sticky prices throughout the exposition). Compared to the previous section, the model encompasses both a corridor and a floor system, allows for the determination of the price level, and analyzes the interactions between fiscal and monetary policies at the financial level.

How will the model differ from the existing literature? In the models of Nagel (2016) and Drechsler, Savov, and Schnabl (2017), the classical dichotomy holds, and the paths of the nominal rate and inflation are independent of real variables and the convenience yield. Since the monetary policy rate set by the central bank coincides with the illiquid nominal rate governing intertemporal consumption, the system of three equations that includes the Euler equation, the Taylor rule (with an inflation coefficient sufficiently large), and the endowment process fully determines the illiquid nominal rate, the price level, and the real rate. The

convenience yield only sets the nominal yield on the T-bill, and it is entirely determined by the real supply of deposits and T-bills, assuming their nominal supply is exogenous. In this setting, note that any exogenous monetary policy tightening will always increase the convenience yield, as the decline in reserve supply needed in a corridor system to raise the illiquid rate increases the liquidity premium. Additionally, it cannot address a floor system because the significant increase in reserve supply required to implement it would imply that the IOR and the T-bill yield rise to equal the monetary policy rate, and the liquidity premium would always be zero for all assets.

In my model, the monetary policy rate set by the central bank does not coincide with the nominal rate in the Euler equation, as in Canzoneri and Diba (2005), Lenel, Piazzesi, and Schneider (2019), Piazzesi, Rogers, and Schneider (2021). The central bank sets a rate with a liquidity premium (the rate on liquid reserves). Still, the household decides on intertemporal consumption based on the rate of loans, which are illiquid. Therefore, the nominal rate in the Euler equation differs from the monetary policy rate by an amount equal to the convenience yield on central bank reserves. This convenience yield will never be zero because reserves are always useful as collateral. As will become apparent, this feature is necessary to explain why, under a floor system where the opportunity cost of holding reserves is zero, a monetary policy tightening decreases the convenience yield on T-bills⁷.

The liquidity premium will arise at the bank level because banks choose between bonds and reserves as collateral. At the end of this Section, I discuss its connection with models where the liquidity premium arises at the household level (when households obtain transaction services from either deposits or government bonds).

4.1 Households

Two features are introduced compared to the previous Section. First, the problem is set in nominal terms to allow inflation to impact the real liquidity supply. Second, output remains fixed at $Y_t = Y$, but there will be exogenous shocks to liquidity demand. This will enable analysis of the correlations between the supply of reserves and T-bills across the business cycle.

⁷Canzoneri and Diba (2005) show how the price level is determined in this type of model. Suppose government bonds provide transaction services (either as collateral or as input in the utility function). In that case, the interaction between monetary and fiscal policy determines whether or not the price level is determined. As a result, a much broader class of interest rate rules, even interest rate pegs, can achieve price determinacy.

The representative household seeks to maximize the objective:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [u(C_t) + \chi_t \log(D_t/P_t)] \quad (13)$$

where C_t is consumption, and liquidity services are supplied by money in the form of deposits. D_t/P_t denotes real balances of deposits. As before, these liquidity benefits arise from some unmodeled use in intra-period transactions. P_t is the nominal price of the consumption good. A model period is one quarter, and deposits and T-bills have a one-period maturity.

Deposit demand is subject to an exogenous shock:

$$\log \chi_{t+1} = (1 - \rho_\chi) \log \bar{\chi} + \rho_\chi \log \chi_t + \epsilon_{t+1}^\chi, \quad \epsilon_{t+1}^\chi \sim N(0, \sigma^2) \quad (14)$$

where $\epsilon_{t+1}^\chi > 0$ represents a positive liquidity demand shock (households want to hold more deposits). This could be caused by a flight to safety, unexpected needs for transaction services, etc.

Households optimize subject to the flow budget constraint:

$$\begin{aligned} D_t + k_t A_t - L_t + P_t C_t &= D_{t-1}(1 + i_{t-1}^d) \\ &+ k_{t-1}(A_t + P_t Y_t) - T_t - L_{t-1}(1 + i_{t-1}) + \Omega_t \end{aligned} \quad (15)$$

where all variables are similar to the last Section except for the price level P_t .

The household first-order condition for consumption yields an Euler equation which now accounts for expected inflation:

$$1 = \beta \mathbb{E}_t \left[\frac{u_c(C_{t+1})}{u_c(C_t)} \frac{1 + i_t}{\Pi_{t+1}} \right] \quad (16)$$

where u_c denotes the first derivative of $u(\cdot)$ and $\Pi_{t+1} = P_{t+1}/P_t$. As before, the illiquid rate i_t in Equation (16) governs the household's intertemporal decisions; thus, this is the rate the monetary authority looks to affect. However, as explained below, the central bank's policy rate is the rate on deposits, i_t^d , which will carry a liquidity premium and trade below the i_t .

The first-order condition for real deposit balances is:

$$\frac{\chi_t}{\frac{D_t}{P_t}} = u_c(C_t) \frac{i_t - i_t^d}{1 + i_t} \quad (17)$$

Equation (17) reflects the version in nominal terms of the “household's cost of liquidity” defined in the previous Section. It is also declining in real balances: the marginal benefit of payment instruments is declining in the overall quantity held.

4.2 Banks

I set up the financial intermediaries such that the model encompasses both a corridor and a floor system. The former collapses to the latter as the supply of reserves grows. The model will not account for the endogenous decision to transition from a corridor to a floor system. Instead, I will derive the implications of the model under both systems independently of each other.

The corridor system requires the inclusion of an interbank market with scarce reserves, where heterogeneous banks can trade reserves among themselves at a premium. However, I aim to maintain the tractability of the representative bank framework. To introduce these features within a representative bank, I build on the model of Curdia and Woodford (2011), which relies on a representative bank facing real intermediation and liquidity costs that only reserves can alleviate. In particular, the bank consumes real resources in managing liquidity, $\Psi(d_t, m_t^d)$, $m_t^d \equiv M_t^d/P_t$, where the cost is a positive function of the scale of operations and a decreasing function of central bank reserves held, $\Psi_{dt} > 0$, $\Psi_{mt} < 0$. These costs can be attributed to, for example, the expenses associated with unexpected deposit withdrawals from households. Importantly, $\Psi_{mt} < 0$ means that reserves are more helpful in handling liquidity costs than other assets: if they are sufficiently scarce, banks are willing to pay a higher price. Additionally, $\Psi_{dm} < 0$, meaning that reserves reduce marginal costs for any scale of operations.

I also assume that for any scale of operations, there exists a finite level of reserves $\bar{m}(\cdot)$ for which $\Psi_{mt} = 0$, $\Psi_{dt} = 0$ for all $m_t^d \geq \bar{m}$. The marginal cost of managing deposits, Ψ_{dt} , drops to zero when reserves are plentiful. This assumption resembles the banks' demand for reserves in the models of Poole (1968) and Afonso et al. (2020). This tries to capture that liquidity costs are increasing in the scale of operations (for example, a higher chance of unexpected withdrawals and greater precautionary liquidity demand), decreasing in reserves (for instance, as they meet regulatory requirements and facilitate intraday operations), and that with a sufficiently large amount of reserves, reserve demand by banks becomes flat, $\Psi_{mt} = 0$, once they are above their precautionary demand level, $\bar{m}(l_t)$.

Intuitively, $\Psi(d_t, m_t^d)$ represents the “banks' cost of liquidity”. In a corridor system, this cost is active and helps determine the supply of deposits. Under a corridor system, the central bank issues new reserves and reduces this cost, thereby facilitating the expansion of the deposit supply. Under a floor system, reserves are abundant and this channel is no longer operative.

This model differs from other papers on corridor systems, such as Nagel (2016) and

Drechsler, Savov, and Schnabl (2017), in that the central bank does not set the illiquid rate based on the household's Euler equation. Instead, the interest rate set by the central bank is lower than that of other rates due to a convenience yield on liquid assets.

Banks maximize the present value of their net cash flow, discounted at the illiquid rate, i_t , as before. The new net nominal cash flow is given by:

$$M_{t-1}(1+i_{t-1}^m) - M_t + B_{t-1}^h(1+i_{t-1}^b) - B_t^h + L_{t-1}(1+i_{t-1}) - L_t - D_{t-1}(1+i_{t-1}^d) + D_t - \Psi(m_t, d_t) \quad (18)$$

Banks can issue deposits only if they have sufficient collateral to back them:

$$D_t \leq M_t + \rho_b B_t^h \quad (19)$$

As before, deposits are a cheaper source of funding than equity because they provide nonpecuniary services to households. Without a leverage constraint, it would be optimal to fund the bank entirely with deposits. Therefore, the leverage constraint will always bind in equilibrium.

The first-order conditions for reserves, T-bills, and deposits are, respectively,

$$\frac{i_t - i_t^m}{1 + i_t} = \gamma_t - \Psi_{mt} \quad (20)$$

$$\frac{i_t - i_t^b}{1 + i_t} = \rho_b \gamma_t \quad (21)$$

$$\frac{i_t - i_t^d}{1 + i_t} = \gamma_t + \Psi_{dt} \quad (22)$$

where γ_t is the Lagrange multiplier on the leverage constraint. The left-hand side of Equation (20) is the marginal cost of holding a reserve: the interest differential with a loan. The right-hand side shows the marginal benefit: more highly liquid collateral plus a lower intermediation cost (recall that $\Psi_{mt} < 0$). Similarly, Equation (21) equates the marginal cost of an extra T-bill with its marginal benefit: a less binding leverage constraint. As before, this liquidity premium represents the “bank's cost of collateral” in nominal terms, and it captures that banks value T-bills not only for their interest rate but also for their collateral value, which allows them to issue deposits.

Subtracting the FOC on deposits from the FOC on reserves gives:

$$\frac{i_t^d - i_t^m}{1 + i_t} = -\Psi_{mt} - \Psi_{dt} \quad (23)$$

The rate i_t^d resembles the interbank rate central banks target when implementing a corridor system (as in Curdia and Woodford, 2011). The choice of i_t^d as the central bank's

policy rate can be justified as follows. In the real world, most central banks set the interbank rate at which banks trade reserve balances with each other intraday. The interbank rate can, therefore, be seen as the rate at which banks can obtain liquid funding whenever needed. Although there is no interbank market in the model, the policy rate i_t^d captures the same idea: it is the rate at which banks can obtain liquid funding, but, in this case, from households⁸. $i_t^d - i_t^m$ in Equation (20) captures the bank’s marginal “cost of liquidity”. For $m_t^d < \bar{m}_t$ and a given i_t^m , Equation (23) delivers a one-to-one relationship between the supply of reserves and the interbank rate, i_t^d . For a given i_t^m , an increase in the supply of reserves reduces the liquidity cost of banks (right-hand side of the equation) and thus reduces the premium on reserves, reducing i_t^d (in the left-hand-side). As explained before, if $m_t^d \geq \bar{m}_t$, then $\Psi_m = \Psi_d = 0$, banks face no liquidity cost, and the interbank rate equals the interest paid on reserves, $i_t^d = i_t^m$. Intuitively, since reserves are abundant, banks no longer pay a premium on them.

The monetary policy rate set by the central bank will be either i_t^d (corridor system) or the IOR, i_t^m (see the following two subsections for more details). The banks’ problem helps to see why the two key features of the model are important, e.g., the leverage constraint and the fact that the monetary policy rate must be the IOR and not the illiquid rate, i_t . In an alternative version of the model where (1) banks face no leverage constraints (only incur the liquidity shocks Ψ_t) and (2) the central bank sets the illiquid rate i_t instead, which is the one that shows up in the household’s Euler equation, then the significant increase in reserve supply needed to implement the floor system would imply that both i_t^m and i_t^b would rise to equal i_t . In other words, the central bank must always make the liquidity premium on any asset zero. The current setting allows me to keep a positive convenience yield even when reserves are abundant.

A second issue that the first-order conditions clarify is that banks have no market power over deposits. Thus, the rate paid on deposits equals the interest banks receive on reserves. I do not take a stand on the market structure in the deposit market. Still, rather, this should be seen as a simplifying assumption to focus all the attention on the liquidity premium of government debt⁹.

⁸There are alternative ways of modeling an interbank market that give a role to the supply of reserves (see, for example, Arce et al. (2020)), but introducing a more formal interbank market would complicate the model without improving the exposition of the underlying mechanisms.

⁹Drechsler, Savov, and Schnabl (2017) analyze the implication of banks’ market power on the transmission of monetary policy.

4.3 Government and Central Bank

The government is comprised of a fiscal authority and a central bank. The fiscal authority's flow budget constraint is given by:

$$B_t^s = (1 + i_{t-1}^b)B_{t-1}^s + G_t - T_t - T_t^{cb} \quad (24)$$

where B_t^s is total debt issued, G_t is fiscal spending, T_t are tax/transfers to the household, and T_t^{cb} are transfers from the central bank.

The central bank buys government bonds and pays with the issuance of new reserves to the banking system. Its budget constraint is:

$$M_t^s + (1 + i_{t-1}^b)B_{t-1}^{cb} = B_t^{cb} + M_{t-1}^s(1 + i_{t-1}^m) + T_t^{cb} \quad (25)$$

where M_t^s is the supply of reserves and B_t^{cb} are its holdings of government bonds. Notice that $B_t^s = B_t^h + B_t^{cb}$, and therefore the budget constraint for the consolidated public sector is:

$$B_t^h + M_t^s = (1 + i_{t-1}^b)B_{t-1}^h + M_{t-1}^s(1 + i_{t-1}^m) + G_t - T_t \quad (26)$$

Letting $U_t = B_t^h + M_t^s$, the consolidated budget constraint becomes $U_t = U_{t-1} + Z_t$, where $Z_t = i_{t-1}^b B_{t-1}^h + M_{t-1}^s i_{t-1}^m + G_t - T_t$ is the total deficit, inclusive of interest payments.

In real terms, the budget constraint becomes:

$$u_t = \frac{u_{t-1}}{\Pi_t} + z_t \quad (27)$$

where lower case letters denote real variables and $\Pi_t = P_t/P_{t-1}$. I assume that fiscal policy is governed by a rule for setting the real deficit:

$$z_t = ze^{\zeta_t} - \rho_g u_{t-1} \quad (28)$$

where ζ_t will be an exogenous deficit shock and z a parameter to be calibrated. The parameter ρ_g captures the response of the current deficit to the outstanding debt. A positive value resembles a government that follows a sustainable path, as Bohn (1998) describes. For simplicity, I assume $G_t = 0$. In this model, bonds' transaction services will affect equilibrium inflation. Therefore, fiscal and monetary policy will interact in a way absent in standard models where bonds do not provide transaction services. Canzoneri and Diba (2005) show that, in this framework, the government needs to set fiscal policy in real terms to achieve local determinacy in the inflation rate.

4.4 Monetary policy Implementation

The monetary authority conducts monetary policy to meet an interest rate operating target, i_t^{d*} , according to the policy rule

$$i_t^{d*} = \psi \Pi_t^{\psi_\pi} e^{\xi_t} \quad (29)$$

where ξ_t is an exogenous monetary policy shock. Equation (29) rules out the possibility of a zero lower bound, as the calibration for New Zealand in the period before 2020 will not involve such episodes.

In a corridor system, reserves are scarce and are valued as a means to manage liquidity shocks, in addition to addressing leverage constraints. The central bank thus targets i_t^{d*} and elastically supplies reserves to achieve that rate. The quantity of reserves is not an independent policy instrument. If the central bank lowers the supply of reserves but maintains the same interest on reserves (IOR), the interbank rate will rise above the target.

In a floor system, $i_t^{d*} = i_t^m$, and the central bank sets an independent path for the quantity of reserves. I assume the central bank maintains abundant reserves, and banks do not incur liquidity costs. Notice that although their supply is disentangled from the policy rate, it still matters as it affects the collateral supply available to banks.

This represents how the Fed has conducted monetary policy since 2008. The interest rate on reserves and the federal funds rate are the tools used to reach its inflation goal. The supply of reserves is not followed closely¹⁰. Indeed, the Fed’s January 2019 statement reads: “*The Committee intends to **continue** to implement monetary policy **in a regime in which an abundant supply of reserves** ensures that control over the level of the federal funds rate and other short-term interest rates is exercised primarily through the setting of the Federal Reserve’s administered rates, and in which **active management of the supply of reserves is not required***” (emphasis is mine).

Notice that the liquidity premium on reserves drives a wedge between the rate that the central bank controls, i_t^d or i_t^m , and the rate that the central bank wants to affect, which is the intertemporal interest rate governing consumption/savings decisions, i_t . Therefore, the transmission from the policy rate to the intertemporal rate will depend on the endogenous response of convenience yields. This will be explored in more detail in the next Section.

Definition: An equilibrium in this economy are prices $\{i_t, i_t^b, i_t^m, A_t, \Pi_t\}$ and policy rules $\{C_t, d_t, l_t, b_t^h, d_t^s, b_t^s, l_t^s\}$ such that:

¹⁰Except for the period starting in the second half of 2022, when, due to the inflation surge, the Fed decided to run down the size of the balance sheet.

1. Given prices, households maximize lifetime utility, banks maximize future cash flows,
2. All markets clear: $k_t = 1$, $D_t = D_t^s$, $L_t = L_t^s$, $B_t^h + B_t^{cb} = B_t^s$.

The remainder of the paper has a predominantly qualitative goal: to show how fiscal and monetary policy interact in determining the liquidity value of government debt. The model is too simplistic to provide a comprehensive quantitative characterization of the New Zealand economy. However, to ensure that the analysis is based on reasonable values of the parameters, I calibrate the model to match New Zealand's debt levels, as well as the liquidity premium on its assets (see Appendix F).

4.5 Discussion

A typical modeling decision in the literature is whether the liquidity premium arises at the bank or the household level. In my model, the liquidity premium on government bonds arises at the bank level because bonds are good collateral for issuing deposits. The liquidity premium on government debt will equal the spread between the rate on household loans and the rate on government bonds. The liquidity premium represents the cost of collateral to banks, and government bonds compete with reserves as collateral. This aligns with most recent papers on liquidity premia, where banks hold government bonds and earn a convenience yield due to micro-founded balance sheet costs (Lenel, Piazzesi, and Schneider, 2019).

Alternatively, the liquidity premium could arise at the household level, as in Nagel (2016) and Drechsler, Savov, and Schnabl (2017). In this case, both bonds and bank deposits would be in the household's utility function. The liquidity premium on government debt will still equal the spread between the rate on loans and government bonds but with a different interpretation. In this case, the liquidity premium represents the household's cost of liquidity, and government bonds compete with deposits as a source of liquidity.

The two approaches are qualitatively similar regarding the determination of the liquidity premium. For example, in Nagel (2016), banks are a veil and mechanically increase the supply of deposits whenever the central bank increases the supply of reserves. Therefore, the insights obtained in both settings are qualitatively similar.

5 Monetary and Fiscal Policy Interactions

The following three subsections characterize the implications for the liquidity premium of (1) monetary policy shocks, (2) the interaction between the central bank and the fiscal authority

along the business cycle, and (3) shocks to the supply of debt.

5.1 Monetary Policy Shocks

I discuss three implications for the liquidity premium: the response to a monetary policy tightening differs between a floor and a corridor system, I explain the impulse response functions of the relevant variables, and I characterize the implications for the fiscal capacity of the government.

Proposition 1 in Section 3 showed that the liquidity premium responded negatively to an exogenous monetary policy tightening under a floor system. The following Proposition shows that the response is the opposite under a corridor system.

Proposition 2. *An exogenous increase in the monetary policy rate under a corridor system increases the liquidity premium on T-bills.*

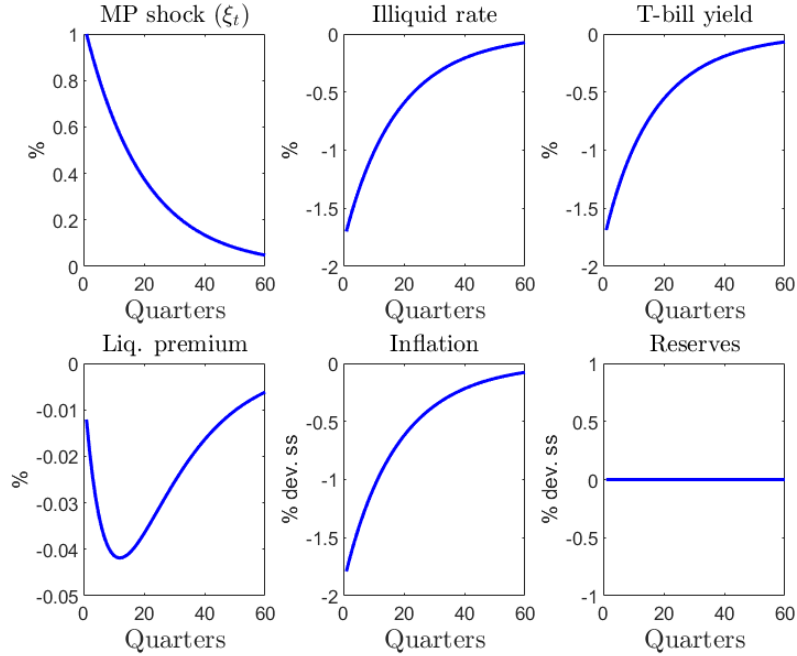
Proof: From the bank's first order conditions, we have that $LP_t \equiv (i_t - i_t^b)/(1 + i_t) = \rho_b \gamma_t$ and thus the response of the liquidity premium to a monetary policy tightening depends entirely on the response of γ_t , $\frac{dLP_t}{di_t^d} = \rho_b \frac{d\gamma_t}{di_t^d}$, where i_t^d is the monetary policy rate under a corridor system. Since γ_t is the Lagrange multiplier on the bank's leverage constraint, its response to policy tightening depends on how M_t (reserve supply) responds to policy changes. Under a corridor system, the central bank actively adjusts the reserve supply to enforce i_t^d . If the policy rate increases, the central bank reduces M_t , which tightens collateral constraints more aggressively, $dM_t/di_t^d < 0$. Therefore $d\gamma_t/di_t^d > 0$. $\square$.

Next, I consider a positive realization of ξ_t in the Taylor rule (29) when the central bank implements a floor system. The monetary policy shock in the model is assumed to follow an AR(1) process with a coefficient of 0.95. The 25 basis point shock increases the interest paid on reserves, with everything else equal, by 1% at annual rates.

The distinctive feature of the floor system is that banks face no liquidity costs and only collateral costs. The monetary policy shock involves a change in the IOR without a change in the supply of collateral (reserves). Banks are now being paid more for the reserves they hold. Therefore, their cost of collateral drops. In this model, banks pass through this lower cost by raising the interest on deposits, lowering the households' liquidity cost.

How do the rest of the variables respond? Since the household's liquidity cost has decreased, there is increased demand for liquid deposits. This lowers the demand and the price of the consumption goods. Thus, inflation and the illiquid rate fall.

Figure 5: Monetary Policy Tightening - Floor System



How does this interact with the fiscal authority? From the perspective of the government, lower inflation increases the current debt burden of the government through the budget constraint, which, absent an adjustment in the primary deficit/surplus, needs to be addressed with an increase in borrowing (see Equation (27)). A rise in the real debt stock leads the Treasury to issue more T-bills, expanding the supply of collateral that backs deposits. This loosens banks' leverage constraint, lowers its shadow price γ_t , and hence reduces the marginal liquidity value. The lower marginal value of liquidity services lowers the liquidity premium of T-bills, as shown in Figure 5. In the end, the response of the government reinforces the original effect of the monetary policy tightening. Alternatively, in a model with endogenous income and sticky prices, the shock would increase i_t^m with little immediate effect on the price level and the real value of government debt. However, it would reduce demand for consumption and, therefore, lower the convenience yield through reduced demand for liquidity.

What are the implications for the fiscal capacity of the government? Standard fiscal capacity analysis indicates that higher interest rates reduce the expected present discounted value of primary surpluses, thereby reducing the sustainable debt-to-GDP ratio. However, the model implies the existence of an additional channel: higher interest rates can reduce the fiscal capacity through their negative effect on the liquidity premium.

To see this, take the government’s budget constraint in (27), ignoring for now the role of reserves as a liability for the government. Iterate it forward, and impose the transversality condition $\lim_{n \rightarrow \infty} E_t[q_{t,t+n}b_{t,t+n}] = 0$ to get:

$$b_t = \mathbb{E}_t \left[\sum_{j=1}^n q_{t,t+j} s_{t+j} \right] + \mathbb{E}_t \left[\sum_{j=1}^n q_{t,t+j} (i_{t+j} - i_{t+j}^b) b_{t+j} \right] \quad (30)$$

where $q_{t,t+j} = \mathbb{E}[\beta^j(u'(C_{t+j})/u'(C_t))]$ is the model-based stochastic discount factor. The left-hand side in Equation (30) corresponds to the current debt-to-GDP ratio (with the steady state endowment normalized to one). The first term on the right-hand side is the expected discounted present value of future primary surpluses. The second term is the expected discounted present value of future liquidity services.

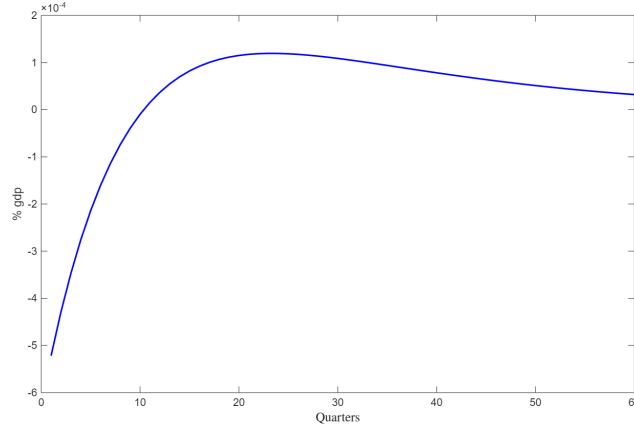
What Equation (30) shows is that the fiscal capacity of the government increases when government debt provides safety and liquidity services to investors: the current value of debt-to-GDP can be above the present discounted value of primary surpluses thanks to the second term on the right-hand side, which depends on the magnitude of the liquidity premium, $i_t - i_t^b$. This term is what Reis (2021) refers to as the “debt revenue” term. Fiscal capacity analysis for the U.S. by Mian, Sufi, and Straub (2022) and Jiang et al. (2022) assumes the debt revenue term depends negatively on the debt supply.

Equation (30) provides an indication of the economic magnitude of the convenience yield in New Zealand’s data. An average liquidity premium of 1.2% over the period, coupled with a mean debt-to-GDP ratio of 0.42, yields a flow debt revenue of 0.5% of GDP. Discounting at the mean illiquid rate of 2.44% (4.44% nominal rate minus 2% inflation) and ignoring risk or uncertainty in calculating the present value over a long time horizon implies that debt revenue can sustain $0.5/0.024 = 20.8\%$ of GDP in public debt. This number is lower than the ones obtained by Reis (2022) for the U.S. and other G7 countries.

A few caveats to this magnitude. First, it assumes that all debt is short-term and, therefore, enjoys the convenience yield of T-bills. A more complete analysis would consider debt at different maturities and consider that longer maturities are less money-like than T-bills. Second, it does not consider central bank reserves as a potential debt obligation for the Treasury.

What is the impact of monetary policy on the fiscal capacity of the government? In the model, a monetary policy shock increases the government’s debt burden in real terms (due to the decline in inflation), which in turn increases borrowing and reduces future deficits, as per the fiscal rule in (28). This impact is qualitatively the same regardless of how monetary policy is conducted.

Figure 6: Impulse Response Function of Debt Revenue to a Monetary Policy Shock



The floor system will have a distinctly negative impact on the “debt revenue” term in (30). Figure 6 below shows the response of the “debt revenue” term to the same monetary policy shock analyzed in Section 5.1. The fiscal capacity of the government drops along with the debt revenue term. The “debt revenue” term falls on impact and it stays below its steady state value for around ten quarters.

This result can provide new insights into the sharp tightening cycle that central banks of advanced economies have initiated since early 2022. This tightening coincided with high debt-to-GDP ratios resulting from the fiscal response to the COVID-19 crisis. Although more research is needed, it might be relevant to the discussion on the optimal size of the central bank’s balance sheet.

5.2 Interactions Along the Business Cycle

A key feature of the model in Section 4 is that the liquidity premium creates a wedge between the rate that the central bank controls and the rate it aims to influence (the intertemporal rate). Therefore, the transmission from the policy rate to the intertemporal rate will depend on the response of the liquidity premium. Since the corridor and the floor systems manage liquidity differently, they will have different outcomes.

For this subsection, I assume that what drives the business cycle are exogenous shocks to households’ demand for liquidity. These will have an impact on interest rates, inflation, and on the supply of reserves and T-bills (via the response of the central bank and the fiscal authority). Analogous results would be obtained if the shocks driving the business cycle were exogenous shocks to the endowment.

Proposition 3. *Under a floor system, the reaction of the convenience yields dampens the*

transmission of changes from the interest on reserves to the intertemporal rate in response to a liquidity demand shock.

Proof: Rewrite the household's first order condition on deposits to get: $1 + i_t^d = (1 + i_t)(1 - (\gamma_t + \Psi_{dt}))$. In response to a liquidity demand shock, $(1 + i_t)$ responds directly from the intertemporal Euler equation, and the previous expression makes clear that the response of i_t^d will differ in magnitude with the response of i_t . The difference will be proportional to the reaction of the convenience yield and liquidity costs. The response of i_t^d depends on $\frac{d\gamma_t}{d\chi_t}$ and $\frac{d\Psi_{dt}}{d\chi_t}$. Since γ_t is the Lagrange multiplier on the bank's leverage constraint and Ψ_{dt} is the marginal cost of deposits, the response of i_t^d to policy tightening depends on how M_t (reserve supply) responds. Take an exogenous decrease in liquidity demand that increases consumption demand and the price level. Under a corridor system, the central bank reduces the supply of reserves, and therefore $d\gamma_t/d\chi_t > 0$ and $d\Psi_{dt}/d\chi_t > 0$ (because we assumed $\Psi_{dm} < 0$). Under a floor system, $1 + i_t^m = (1 + i_t)(1 - \gamma_t)$. There is no change in the supply of reserves, but the hike in the IOR has a marginal impact on the valuation of the collateral constraint, $d\gamma_t/di_t^m < 0$ (see Proposition 1). Therefore, $\frac{\partial i_t^d}{\partial \chi_t} < \frac{\partial i_t^m}{\partial \chi_t}$, i.e., the adjustment needed in i_t^d will be smaller than under a floor system. $\square$

To see the intuition of this result, consider a positive realization of the liquidity demand shock, $\epsilon_{t+1}^X > 0$. This increases the demand for liquid assets and, therefore, a drop in the demand for consumption and in the price of the consumption good. Since output is constant, interest rates need to fall: the fall in inflation calls for lower policy rates and a lower intertemporal rate, i_t , to encourage consumption.

The central bank will accommodate this by reducing the interbank rate (under a corridor system) or the interest on reserves (under a floor system). However, the pass-through of this increase to the intertemporal rate will depend on the response of the households' cost of liquidity (Equation (17)). Under a corridor system, the increase in the supply of reserves needed to lower the interbank rate will translate into a larger supply of deposits (via the binding collateral constraint), thereby accommodating the original shock to liquidity demand and compensating for part of the drop in inflation. The interest on reserves under a floor system will require a more significant decrease in magnitude. The reason is that there is no increase in the supply of reserves, so there is no accommodation of a larger demand for deposits through a corresponding increase in supply. The whole shock needs to be accommodated by the increase in the IOR.

How does this interact with the response of the fiscal authority? From the government's perspective, lower inflation increases the current debt burden through the budget constraint,

necessitating an increase in borrowing in the future (as discussed in the previous subsection). The larger supply of T-bills will increase the supply of high-quality collateral for banks. Under a corridor system, this effect contributes to the larger supply of reserves, and the impact on the supply of deposits is even more significant. This accommodates the larger demand for deposits, the fall in inflation is lower, and the response needed in the policy rate is also lower. Alternatively, in a model with endogenous income and sticky prices, the shock would increase the demand for liquid deposits with little effect on the price level. The government would service this increased demand by issuing more, and the conclusions would be equivalent.

Importantly, this indicates that the response of the central bank and the fiscal authority yields a positive correlation between the supply of T-bills and the supply of reserves in a corridor system, aligning with the empirical facts presented in Section 2.

Under a floor system, on the other hand, the larger supply of T-bills increases the supply of high-quality collateral, but with no accompanying effect of reserves, and thus, the effect is more muted. Still, the response needed in the IOR is larger than for the policy rate under the corridor system¹¹. This also indicates that the response of the central bank and the fiscal authority yields a non-significant correlation between the supply of T-bills and the supply of reserves in a floor system, aligning again with the empirical facts presented in Section 2.

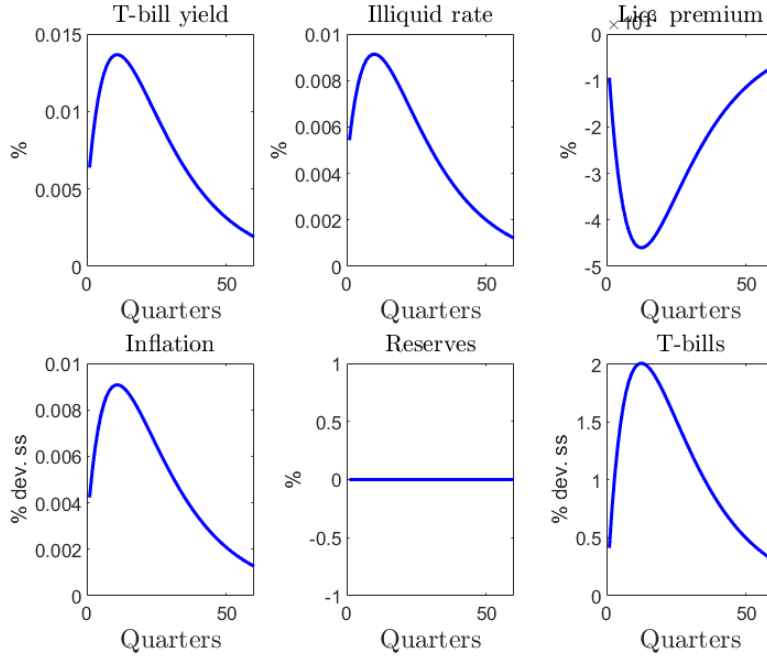
One implication of this result is that a central bank will need to adjust its response function to business cycle shocks more under a floor system than under a corridor system. A central bank transitioning from a corridor to a floor system without considering this issue risks persistent inflation above its target.

5.3 Debt Supply Shocks

I consider an exogenous increase in the deficit financed by the rise in the supply of government debt. The central bank implements a floor system. In New Zealand data, the annual growth rate of the supply of T-bills between January 2008 and February 2020 has a mean of 6.25% and a standard deviation of 79. The debt issuance in response to the global financial crisis influences this sizable standard deviation. I consider an increase in ζ_t of 2% over a year. The blue line in Figure 7 compares the impulse responses of the yield on T-bills, the liquidity premium, inflation, and central bank reserves for the calibration described in the Appendix F.

¹¹The implication that the proportional magnitude of the response of i_t is different in a corridor versus a floor system can also be found in the New-Keynesian model of Piazzesi, Rogers, and Schneider (2021).

Figure 7: Positive Debt Supply Shock - Floor System



The exogenous increase in the deficit is financed by an increase in government borrowing, which directly affects the liquidity premium as the debt supply rises. Thus, the marginal convenience yields of liquid assets decrease. In addition, the increase in the supply of collateral loosens banks' leverage constraint, increases nominal deposits, and thereby puts upward pressure on the price level (via the deposits first-order condition), which calls for an increase in i_t^m via the central bank's rule. Notice that the rise in i_t^m does not imply a decrease in the supply of reserves, which is the main feature of the floor system. Intuitively, from a liquidity preference perspective, the yields on liquid assets have to rise to encourage banks to hold the new supply of T-bills, and a reduction in the convenience yield accommodates this.

Under a corridor system, the central bank would also raise the interest rate (the interbank rate in this case), which would need a reduction in reserves. The drop in reserve supply could offset the increase in government debt and thus have a null effect on the overall supply of liquid assets and no effect on the liquidity premium.

This is what Nagel (2016) highlights in his results. In his model with a corridor system, bonds and deposits provide transaction services for households (not banks), and thus the liquidity premium arises at the household level. The drop in the supply of reserves has a mechanic impact on deposits, and they offset the increase in debt supply; if bonds and deposits are perfect substitutes, then there would be no change in the overall liquidity supply

(deposits plus bonds) and no change in the liquidity premium. Nagel (2016) contends that the fact that debt supply has no statistically significant effect on the liquidity premium of government debt allows him to conclude that bonds and deposits are perfect substitutes.

One implication from Figure 7 is that, under a floor system, the debt supply impacts the liquidity premium even when reserves and bonds are perfect substitutes (they are perfect substitutes in the collateral constraint in Equation (19)). d’Avernas and Vandeweyer (2024) extend the regressions in Nagel (2016) and find that the debt supply has become statistically significant in explaining the liquidity premium of government debt after 2008. Starting from the results of Nagel (2016), they conclude that it must be because deposits and T-bills are no longer perfect substitutes. Figure 7 shows that such a result might not be conclusive about the degree of substitutability between T-bills and other forms of liquidity.

6 Conclusions

This paper has provided new insights into how monetary and fiscal policy interact in determining the liquidity premium of short-term government debt. Clearing up the collinearity between the supply of T-bills and its close substitutes showed that the supply of government debt negatively impacts its liquidity premium. This result mainly implies that a government that borrows too much can reduce the premium and eventually exhaust it. From the fiscal policy perspective, too much borrowing can lessen the debt revenue term in the government’s budget constraint. This confirms previous assumptions in the literature that a government cannot indefinitely exploit the seignorage revenue from the safety/liquidity of its debt.

At the same time, the floor system allows one to estimate the impact of interest rates, stripped of their link to the supply of central bank reserves. The interest on reserves has a negative impact on the liquidity premium, even when central bank reserves have no opportunity cost for banks. This implies that increases in the monetary policy rate under a floor system can further adversely affect the government’s fiscal capacity. As the number of advanced economies implementing an abundant-reserves system has grown in the past decade, it is even more critical to understand how this new system has changed this and other linkages between monetary and fiscal policy. One fruitful avenue for future research will be to consider the endogenous debt maturity decisions by the budgetary authority, considering that long-term bonds have less liquidity value than T-bills.

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Appendix A Data sources

Yields: the interest on reserves, the 3-month bank bill rate, and the 6-month term deposit rate are from the Reserve Bank of New Zealand (RBNZ) statistics. See Wholesale interest rates, table B2 monthly, for the OCR and the bank bill, and Retail interest rates, table B3, for the term deposit rate. The 3- and 6-month yields on T-bills are calculated from auction results published by the Debt Management Office of New Zealand’s Treasury. When more than one instrument is offered, I take the weighted average of successful yields.

Supply of T-bills and deposits: the amount of T-bills and bonds (long-term) outstanding is published monthly by the Debt Management Office of New Zealand’s Treasury. Series are at market value. The central bank’s supply of reserves is obtained from the RBNZ’s balance sheet (see statistics, table R2). The two proxies for the supply of deposits correspond to households’ deposits at registered banks, available at the RBNZ’s statistics (see Household balance sheet, table C22), and total deposits calculated as broad money (Table C50) minus currency in circulation (from the RBNZ’s balance sheet, Table R2).

All these variables are scaled by GDP and are available at FRED (current price Gross Domestic Product in New Zealand, New Zealand dollars, Quarterly). The series was brought to monthly frequency through interpolation.

Other: the VIX index and the U.S. federal funds rate were obtained from FRED (series VIXCLS and EFFR). Data for bank bills’ futures contracts prices (the instrumental variable for the IOR) was obtained from Bloomberg, tickers ZB1 and ZB2. Supply for U.S. T-bills was obtained from TreasuryDirect. U.S. deposit supply corresponds to the sum of checking and savings accounts in Table 4 in the H.6 release (see FRED).

Appendix B Monetary policy in New Zealand

This section relies on Keister et al. (2008) and Selgin (2018) to describe New Zealand’s transition to a floor system. From 1999 until July 2006, the Reserve Bank of New Zealand (RBNZ) implemented a symmetric corridor system in which the benchmark policy rate, the Official Cash Rate (OCR), was kept 25 basis points above the rate paid on reserves and 25 points below what was charged for overnight loans. Because banks’ overnight settlement balances (i.e., reserves) bore an opportunity cost in such a regime, banks held very few such balances, relying instead on intraday credits from the RBNZ to meet their ongoing settlement needs.

Along with its corridor system, the RBNZ implemented a Real Time Gross Settlement

(RTGS) system for wholesale payments, in which interbank payments are settled bilaterally and immediately, thereby becoming final and irrevocable as transactions are processed, rather than at the end of the business day only. This allows payments to be made final as soon as they are processed. If a bank faces any real-time liquidity need in the remainder of the day, it relies on credit from the RBNZ instead of waiting to settle its net balances at the end of each day.

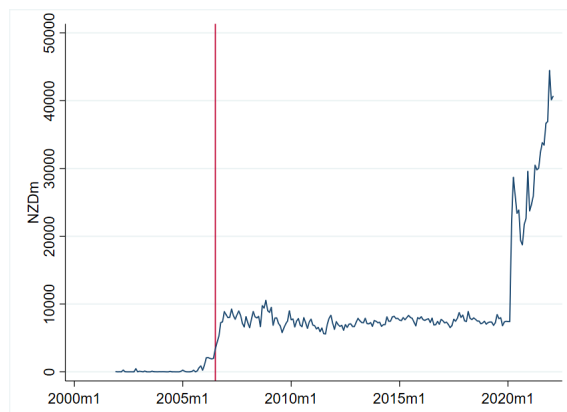
The disadvantage of the RTGS is that it exposes the central bank to credit risk. The RBNZ chose to supply intraday credit through fully secured reverse repo agreements to avoid that risk. However, the Government of New Zealand had been running a fiscal surplus for a few years, and government bonds had become scarce. Since these bonds were the preferred collateral, the scarcity produced broad volatility in the interbank market and often resulted in rates significantly above the target desired to implement monetary policy. The RBNZ had to accept municipal and corporate paper as collateral, which did not eliminate the risk of losses.

The RBNZ reviewed its liquidity management regime in 2005 and announced a new system in early 2006. Recognizing the danger of losses, the RBNZ encouraged banks to rely on overnight settlement balances (reserves) instead of intraday repos to meet their settlement needs. With that end in mind, in July 2006, the RBNZ began its program of “cashing up” the banking system. The first step was to create an additional NZ\$7 billion reserves between July and October. Concurrently, it increased the interest paid on those reserves by 25 basis points (made in five-point increments) to encourage banks to hold them. In addition, the RBNZ stopped providing intraday repos. In the end, the RBNZ achieved the two critical components of a floor system: the interest rate on reserves was equal to the policy rate, and banks were well supplied, if not satiated, with liquidity.

Figure 8 plots the evolution of the supply of reserves. The vertical red line marks the official start of the floor system. The level of reserves went from a negligible amount (NZ\$20 million approx.) to around NZ\$8 billion. This level remained steady until March 2020, when QE raised the supply to almost NZ\$30 billion.

Can we be sure to rule out any endogenous variation in reserve supply? There are two episodes in which the RBNZ injected further liquidity into the banking system, both during the financial turmoil of 2008 (in August 2007 and the Fall of 2009). These are the most likely to be driven by liquidity concerns and thus be endogenous. Figure 9 provides a closer inspection of the evolution of reserve supply. Although the RBNZ occasionally found it desirable to inject some extra reserves into the banking system -as in the two episodes

Figure 8: Reserve supply in New Zealand



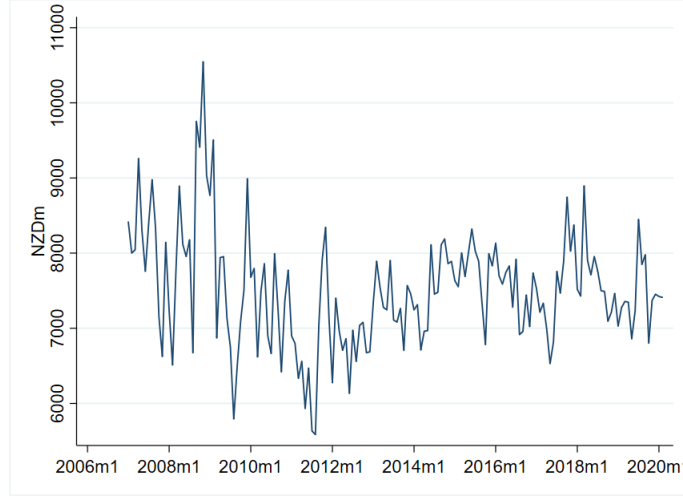
mentioned- those cash additions were modest and transitory. As an extra precaution, I add the supply of reserves or deposits in the regressions to control for any out-of-order variation in their quantities.

Finally, one crucial reform happened in August 2007. A common issue with supplying too many reserves is that banks no longer need to rely on one another to access reserves, effectively ending active trading in the interbank market. This has happened in the US since 2008 in the federal funds market, where the number of transactions has been minimal for the past decade. The RBNZ concluded that it was a good idea to avoid this outcome, as an active interbank market provides information on the health of the banks. However, once the floor system was up and running, it became clear that it encouraged some banks to hoard reserves, accumulating them without incurring any opportunity cost and, thus, without the need to go to the interbank market.

The solution to the reserve hoarding problem was a “tiering system”, with reserves up to a bank’s assigned tier limit earning the interest on reserves and reserves beyond that level earning 100 basis points less. The tier levels were based on banks’ apparent settlement needs but collectively still amounted to the aggregate target of NZ\$7 billion. The overnight interbank lending market remained active thanks to this switch to a tiered system. Banks continued to rely upon one another as lenders of first resort, turning to the RBNZ for overnight funds only as a last resort.

It is important to notice that this tiered system does not affect my identification strategy. Even though an active interbank market might suggest that banks are not fully satiated with liquidity, my strategy only needs the overall supply of reserves not to be correlated with the monetary policy rate, unlike in a scarce-reserve system. Figure 3 shows that this is the case. In contrast, the same graph for a corridor system -like the one in the US before 2008-

Figure 9: Reserve supply in New Zealand: Jul. 2006 - Feb. 2020



would show a significant negative correlation between the federal funds rate and the supply of reserves/deposits.

Appendix C First-Stage Regressions (Section 2)

Appendix D Robustness for Results in Section 2

Table 4 estimates equation (1) with data for New Zealand from July 2006 until July 2021. The dependent variable is the spread between the 3-month bank bill and the 3-month T-bill. I differentiate the dependent and independent variables to avoid concerns with stochastic trends. This avoids spurious correlations between the levels of these variables by removing random walk components in these series. All columns are estimated using OLS, with the standard errors adjusted for autocorrelation using the Newey West estimator.

In Column 1, the coefficient on the supply of T-bills is negative and significant. The coefficient on the IOR is negative, although not statistically significant. Column 2 adds the supply of reserves as a percentage of GDP, which serves as a robustness check to the intuition that under a floor system, reserve supply is exogenous in determining the liquidity premium. The estimation shows that it does not have a significant effect, and the coefficient on the supply of T-bills is similar in magnitude and significance.

In columns 3 and 4, I add the current and lagged values of the independent variables. I do this to consider the possibility that the coefficients in columns 1 and 2 only capture a transitory effect on the liquidity premium. This is especially relevant in this case where I

Table 2: First-Stage Regressions in 2sls Results

	(1)	(2)	(3)	(4)	(5)	(6)
	ior_t	$\ln Tbill/GDP_t$	$\ln HHdeps/GDP_t$	$\ln Debt/GDP_t$	$\ln Deps/GDP_t$	$\ln Res/GDP_t$
$futures_{t-2}$	0.931 (0.015)	0.286 (0.042)	-0.069 (0.006)	-0.115 (0.018)	-0.053 (0.004)	-0.194 (0.018)
$(debt/GDP)_t$	-2.694 (1.647)	6.056 (4.552)	-1.513 (0.669)			
$(debt/GDP)_t^2$	2.655 (1.776)	-9.324 (4.909)	2.142 (0.721)			
$(debt/GDP)_t^3$	-0.760 (0.604)	4.057 (1.669)	-0.831 (0.245)			
$(debt/GDP)_{t-1}$				-0.880 (1.317)	-0.276 (0.262)	0.216 (1.301)
$(debt/GDP)_{t-1}^2$				-7.874 (11.09)	2.013 (2.208)	6.347 (10.96)
$(debt/GDP)_{t-1}^3$				77.11 (162.9)	35.45 (32.44)	79.56 (160.9)
$(tbill/GDP)_{t-1}$	-0.064 (0.141)	0.657 (0.390)	0.059 (0.057)			
$(tbill/GDP)_{t-1}^2$	0.819 (0.955)	1.149 (2.640)	0.755 (0.388)			
$(tbill/GDP)_{t-1}^3$	-0.770 (0.989)	-0.905 (2.735)	-0.796 (0.402)			
$(tbill/GDP)_t$				28.58 (6.471)	5.483 (1.288)	41.55 (6.392)
$(tbill/GDP)_t^2$				-277.3 (65.20)	-56.74 (12.98)	-299.6 (64.41)
$(tbill/GDP)_t^3$				499.9 (200.7)	149.1 (39.96)	667.7 (198.3)
vix	-0.009 (0.002)	0.189 (0.005)	-0.002 (0.001)	-0.008 (0.003)	0.001 (0.000)	0.014 (0.002)
Constant	0.829 (0.493)	-4.409 (1.364)	1.139 (0.200)	-0.522 (0.198)	1.289 (0.039)	-3.450 (0.195)
F-stat	775.6	15.56	45.96	22.60	87.44	46.74
Prob>F	0.000	0.000	0.000	0.000	0.000	0.000
Adj. R^2	0.977	0.440	0.709	0.539	0.824	0.712
Observations	149	149	149	149	149	149

Notes: first-stage regressions for 2sls results in Section 2 in the main text.

Table 3: New Zealand: Determinants of Liquidity Premium - OLS

	Sample: 2006m7-2021m7				Sample: 2006m7-2020m2			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δior_t	-0.083 (0.052)	-0.084 (0.053)	-0.187*** (0.063)	-0.187*** (0.063)	-0.094* (0.056)	-0.101* (0.054)	-0.206*** (0.069)	-0.227*** (0.070)
Δior_{t-1}			0.265*** (0.076)	0.269*** (0.075)			0.273*** (0.077)	0.295*** (0.081)
Δvix_t	0.004*** (0.001)	0.003 (0.002)	0.002 (0.002)	0.000 (0.002)	0.005** (0.002)	0.005** (0.002)	0.005 (0.002)	0.004* (0.002)
Δvix_{t-1}			0.001 (0.004)	0.001 (0.005)			-0.001 (0.006)	-0.001 (0.006)
$\Delta \log(\frac{Tbill}{GDP})_t$	-0.278** (0.124)	-0.290** (0.119)	-0.193** (0.092)	-0.195** (0.077)	-0.371** (0.188)	-0.349* (0.184)	-0.299** (0.117)	-0.284*** (0.108)
$\Delta \log(\frac{Tbill}{GDP})_{t-1}$			-0.048 (0.090)	-0.042 (0.089)			-0.068 (0.119)	-0.054 (0.109)
$\Delta \log(\frac{Res}{GDP})_t$		0.119 (0.086)		0.139 (0.114)		0.180 (0.135)		0.218 (0.145)
$\Delta \log(\frac{Res}{GDP})_{t-1}$				-0.009 (0.080)				-0.129 (0.093)
Constant	-0.006 (0.008)	-0.009 (0.008)	-0.000 (0.006)	-0.002 (0.006)	-0.009 (0.010)	-0.009 (0.010)	-0.003 (0.007)	-0.003 (0.007)
Observations	181	181	181	181	164	164	164	164

Notes: Newey-West standard errors in parentheses (6 lags). The dependent variable is the first difference of the 3-month Bank bill/T-bill spread, measured in percentage points. Independent variables are the same as in Table 1 in the main text.

am using 1-month differences. It can be the case, for example, that new debt issuance by the government has a short-lived impact that fades away soon after. Results show that the supply of T-bills still has a negative and significant impact. Importantly, unlike the results in Nagel (2016), the lagged value is insignificant, suggesting it has a more persistent effect. The results for the IOR confirm that it has a significant negative impact.

In columns 5-8, I replicate the first four columns but with the sample ending in February 2020. Recall that the RBNZ embarked on large-scale asset purchases after March 2020. Large-scale asset purchases involve purchasing assets in exchange for newly issued reserves. Therefore, we should no longer expect the supply of reserves to be exogenous in this situation. The estimation in columns 5-8 is a robustness check as it should be free of any effect of QE

and the possible endogeneity of the supply of reserves. The results show that the impact of the supply of T-bills and the IOR is still significant.

Appendix E Regressions with U.S. Data After 2008

Does the negative effect of the interest on reserves hold in data for other countries? The following Table shows regressions of the liquidity premium for the U.S. after November 2008. The dependent variable is the spread of 3-month AA financial commercial paper and 3-month T-bills. Data for T-bill and reserve supply are from TreasuryDirect and FRED, respectively. The caveats pointed out in the main text still apply, mainly that the supply of reserves is likely to be endogenous to the different QE programs carried out by the Federal Reserve between 2009-2014. However, it is interesting to see the coefficients for the IOR and the supply of T-bills are of the same sign as in Table 4. In particular, both the interest on reserves and the supply of T-bills have a negative effect on the liquidity premium. These results hold when using IVs (not shown).

Appendix F Calibration of the Model

The model is calibrated with data for New Zealand at a quarterly frequency, using the sample 2006:III to 2021:III. Panel A of Table 5 shows the parameters calibrated externally. I pick a value of risk aversion of 1.5 (I assume CRRA utility for consumption) and assume a Taylor rule parameter for inflation of 1.5, as it is common in the literature. As explained in the main text, the parameter ρ_g captures the response of the current deficit to the outstanding debt in the government's fiscal rule. A positive value resembles a government that follows a sustainable path, as Bohn (1998) describes (larger outstanding debt is met with lower deficits, see Equation (28)). The value of $\rho_g = 0.1$ is consistent with a very low deficit, consistent with the sustainable fiscal path shown by New Zealand over the time period.

The rest of the parameters are calibrated under a deterministic steady state. Table 5, Panel B, shows the values for the five parameters. The targets are the illiquid rate, for which I use the yield on 6-month term deposits, the average yield on T-bills, and the OCR during the sample period. I complement this with government debt to GDP and the deficit to GDP. I define a supply of government debt to GDP of 0.42 (OECD data, General Government debt).

The calibration exercise gives a discount factor of 0.994, which is somewhat lower than

Table 4: U.S.: Determinants of liquidity premium, 2008m11-2020m6 - OLS

	(1)	(2)	(3)	(4)
Δioer_t	-0.363*** (0.124)	-0.604*** (0.158)	-0.361** (0.157)	-0.511*** (0.124)
Δioer_{t-1}			0.449*** (0.131)	0.403*** (0.103)
Δvix_t	0.013*** (0.003)	0.012*** (0.002)	0.011*** (0.003)	0.008*** (0.002)
Δvix_{t-1}			0.006*** (0.002)	0.005** (0.002)
$\Delta \log(\frac{Tbill}{GDP})_t$	-0.895*** (0.223)	-0.755*** (0.195)	-0.498*** (0.139)	-0.625*** (0.154)
$\Delta \log(\frac{Tbill}{GDP})_{t-1}$			0.266 (0.167)	0.371*** (0.123)
$\Delta \log(\frac{Res}{GDP})_t$		-0.932*** (0.343)		0.075 (0.167)
$\Delta \log(\frac{Res}{GDP})_{t-1}$				-0.495*** (0.169)
Constant	-0.0127 (0.014)	-0.002 (0.009)	-0.004 (0.005)	0.000 (0.005)
Observations	140	140	139	139

Notes: Newey-West standard errors in parentheses (6 lags). Units are in percentage points. The dependent variable is the first difference of the 3-month AA financial commercial paper/T-bill spread.

usual. However, this difference accounts for the illiquidity of the intertemporal rate in the model. The liquidity preference in the household's utility function, $\bar{\chi}$, takes the value of 3.6365e-04, and the value of T-bills as collateral, ρ_b , equals 1.85. This implies that T-bills are better collateral than reserves. This is consistent with that, unlike reserves, T-bills can be used as collateral for repo transactions.

Table 6 shows how the model matches the targets. The last row computes the liquidity premium on government debt.

Table 5: Calibration

A. Externally calibrated			
	Description	Value	Source
γ	Risk aversion	1.5	Standard
ψ_π	Taylor rule	1.5	Standard
ρ_g	Gov's rule	0.1	Bohn (1998)
B. Calibrated			
	Description	Value	Target
β	Discount factor	0.994	$\text{mean}(i_t)$
$\bar{\chi}$	Liq. preference	3.6365e-04	$\text{mean}(i_t^b)$
ρ_b	Collateral constraint	1.85	$\text{mean}(i_t^m)$
z	Gov's deficit	0.05	Tbill/GDP

Notes: parameters are chosen simultaneously to match the targets.

Table 6: Model moments and Targets (Annualized)

Variable	Description	Data	Model
i_t	Illiquid rate	4.44%	4.44%
i_t^b	Debt yield	3.25%	3.25%
i_t^m	IOR	3.34%	3.79%
b_t	Gov debt/GDP	0.42	0.47
$i_t - i_t^b$	Liquidity premium	1.19%	1.19%

Notes: